PROJECT - Netflix Data Cleaning, Analysis & Visualization

Project Description:

The goal of this project is to perform comprehensive data cleaning, exploratory data analysis (EDA), and visualization on the Netflix content catalog. By transforming raw data into a structured analytical dataset, the project aims to uncover meaningful insights related to strategic content acquisition, audience preferences, and global production trends. These insights will support Netflix in making informed, data-driven decisions to enhance its content strategy, improve viewer engagement, and strengthen its competitive advantage in the streaming industry.

Tech Stack Used:

Microsoft Excel: For dataset view

Python: For Data cleaning & preprocessing, EDA, data visualization

Tableau: For dashboard creation

Dataset Overview:

Netflix is a popular streaming service that offers a vast catalog of movies, TV shows, and original contents. The dataset consist of contents added to Netflix from 2008 to 2021. The oldest content is as old as 1925 and the newest as 2021. This dataset will be cleaned with Python and visualized with Tableau.

Column Descriptions:

Column Name	Description
show_id	Unique identifier assigned to each title in the dataset.
type	Indicates whether the content is a Movie or a TV Show.
title	Name of the movie or TV show.
director	Name of the director of the content.

Column Name	Description
country	Country where the content was produced.
date_added	Date when the title was added to the platform.
release_year	Year the movie or TV show was originally released.
rating	Content maturity rating.
duration	For Movies: duration in minutes; For TV Shows: number of seasons.
listed_in	Genres of the content.

Steps:

1. Importing Necessary Libraries
2. Data Collection & Data Understanding
3. Data Cleaning & Preprocessing
4. Exploratory Data Analysis (EDA)

Below are the steps followed and python codes for doing the analysis.

1. Importing Necessary Libraries

```
#Importing necessary libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud
```

2. Data Collection & Data Understanding

```
#Loading the dataset

data=pd.read_csv("netflix1.csv")
data.head(5)
```

```
  show_id     type                          title         director  \
0      s1    Movie             Dick Johnson Is Dead  Kirsten Johnson
1      s3  TV Show                        Ganglands  Julien Leclercq
2      s6  TV Show                     Midnight Mass    Mike Flanagan
3     s14    Movie  Confessions of an Invisible Girl    Bruno Garotti
4      s8    Movie                        Sankofa     Haile Gerima

        country date_added  release_year rating  duration  \
0  United States  9/25/2021         2020  PG-13    90 min
1        France  9/24/2021         2021  TV-MA  1 Season
2  United States  9/24/2021         2021  TV-MA  1 Season
3        Brazil  9/22/2021         2021  TV-PG    91 min
4  United States  9/24/2021         1993  TV-MA   125 min

                                   listed_in
0                                Documentaries
1  Crime TV Shows, International TV Shows, TV Act...
2                TV Dramas, TV Horror, TV Mysteries
3                Children & Family Movies, Comedies
4   Dramas, Independent Movies, International Movies
```

data.shape

(8790, 10)

There are 8790 rows and 10 columns in the dataset.

data.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8790 entries, 0 to 8789
Data columns (total 10 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8790 non-null   object
 1   type          8790 non-null   object
 2   title         8790 non-null   object
 3   director      8790 non-null   object
 4   country       8790 non-null   object
 5   date_added    8790 non-null   object
 6   release_year  8790 non-null   int64
 7   rating        8790 non-null   object
 8   duration      8790 non-null   object
 9   listed_in     8790 non-null   object
```

```
dtypes: int64(1), object(9)
memory usage: 686.8+ KB
```

```
data.describe()
```

```
       release_year
count   8790.000000
mean    2014.183163
std        8.825466
min     1925.000000
25%     2013.000000
50%     2017.000000
75%     2019.000000
max     2021.000000
```

```
data.dtypes
```

```
show_id        object
type           object
title          object
director       object
country        object
date_added     object
release_year   int64
rating         object
duration       object
listed_in      object
dtype: object
```

3. Data Cleaning & Preprocessing

```
# Convert 'date_added' to datetime
data['date_added'] = pd.to_datetime(data['date_added'])
```

```
data.dtypes
```

```
show_id               object
type                  object
title                 object
director              object
country               object
date_added      datetime64[ns]
release_year            int64
rating                object
duration              object
```

```
listed_in             object
dtype: object
```

```
#checking if there any missing values and duplicate rows
print('\n--- Missing values per column ---')
print(data.isnull().sum())
print('\n--- Duplicate rows count ---')
print(data.duplicated().sum())
```

```
--- Missing values per column ---
show_id        0
type           0
title          0
director       0
country        0
date_added     0
release_year   0
rating         0
duration       0
listed_in      0
dtype: int64

--- Duplicate rows count ---
0
```

There are no missing values and duplicated data.

```
#Feature Engineering
#Extract day, month and year from date_added column
data['year_added']=data['date_added'].dt.year
data['month_added']=data['date_added'].dt.month
data['day_added']=data['date_added'].dt.day

# Extract the numeric part of the duration column
data['duration_int'] = data['duration'].str.extract('(\d+)').astype(int)

# Rename 'duration' to 'duration_text' for clarity (keeping it for now)
data.rename(columns={'duration': 'duration_text'}, inplace=True)

#  Check the result of the duration column cleaning
print("\nDataFrame Head after cleaning:")
print(data[['type', 'date_added', 'year_added', 'month_added', 'duration_int', 'duration_
text']].head())
```

```
print("\nValue Counts for type and duration_int:")
print(data.groupby('type')['duration_int'].describe())
```

```
DataFrame Head after cleaning:
     type date_added  year_added  month_added  duration_int duration_text
0   Movie 2021-09-25       2021           9           90       90 min
1  TV Show 2021-09-24       2021           9            1     1 Season
2  TV Show 2021-09-24       2021           9            1     1 Season
3   Movie 2021-09-22       2021           9           91       91 min
4   Movie 2021-09-24       2021           9          125      125 min
```

```
Value Counts for type and duration_int:
        count      mean       std  min   25%   50%    75%    max
type
Movie    6126.0  99.584884  28.283225  3.0  87.0  98.0  114.0  312.0
TV Show  2664.0   1.751877   1.550622  1.0   1.0   1.0    2.0   17.0
```

For movies, the longest movie has a duration of 312 minutes and the duration of the shortest movie is 3 minutes.

For TV Shows, the longest TV Show has a duration of 17 seasons and the duration of the shortest TV Show is 1 season.

```
# Split the comma-separated genres and count individual genres
#  Split 'listed_in' and explode the DataFrame
genres_data = data['listed_in'].str.split(', ', expand=True).stack().reset_index(level=1, dr
op=True).to_frame('genre')

genres_data
```

```
                genre
0          Documentaries
1         Crime TV Shows
1   International TV Shows
1    TV Action & Adventure
2             TV Dramas
...                 ...
8787  International TV Shows
8787      Romantic TV Shows
8787            TV Dramas
8788             Kids' TV
8789             Kids' TV
```

[19294 rows x 1 columns]

Data cleaning and preprocessing is completed. Next we are going to conduct EDA.

4. Exploratory Data Analysis

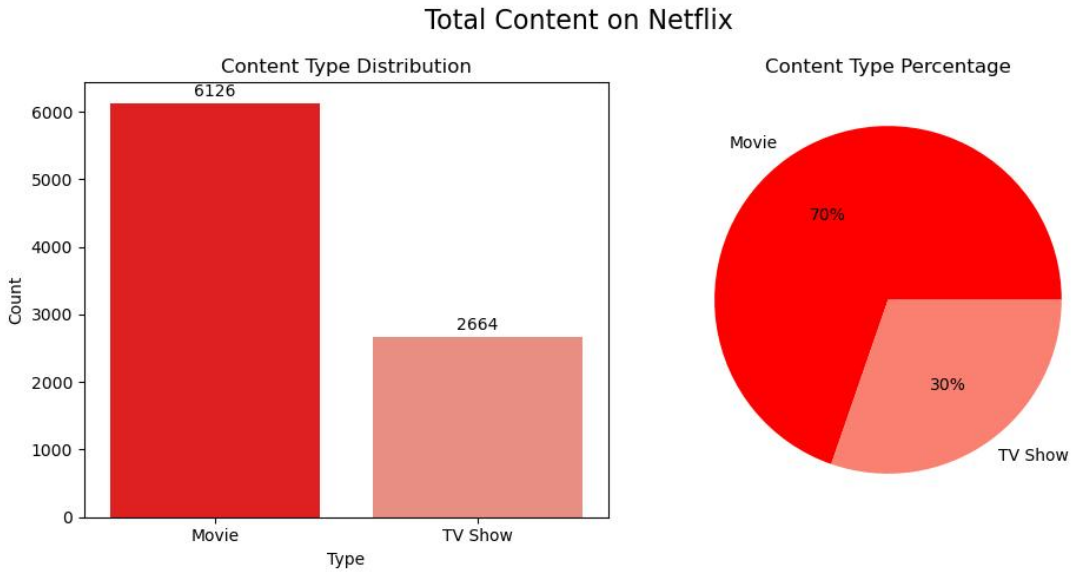
There are the details of 8790 content titles including both movies and TV shows on Netflix in the dataset.

1) Content Type Distribution on Netflix

```
data['type'].value_counts()
```

```
Movie      6126
TV Show    2664
Name: type, dtype: int64
```

```
freq = data['type'].value_counts()
colors = ['red', 'salmon']
fig, axes = plt.subplots(1, 2, figsize=(10, 5))

barplot = sns.countplot(x=data['type'], ax=axes[0], palette=colors)

axes[0].set_title("Content Type Distribution")
axes[0].set_xlabel("Type")
axes[0].set_ylabel("Count")
axes[0].grid(False)

for p in barplot.patches:
    height = p.get_height()
    axes[0].text(p.get_x() + p.get_width()/2,
                 height + 50,
                 f'{height}',
                 ha='center', va='bottom', fontsize=10)

axes[1].pie(freq,
            labels=freq.index,
            autopct='%.0f%%',
            colors=colors)

axes[1].set_title("Content Type Percentage")

plt.suptitle("Total Content on Netflix", fontsize=16)
plt.tight_layout()
plt.show()
```



Findings:

Approximately 70% of the content available on Netflix consists of movies, while the remaining 30% is made up of TV shows. This indicates that Netflix's library is still heavily focused on movie content, although TV shows also form a significant portion of the platform's offerings.

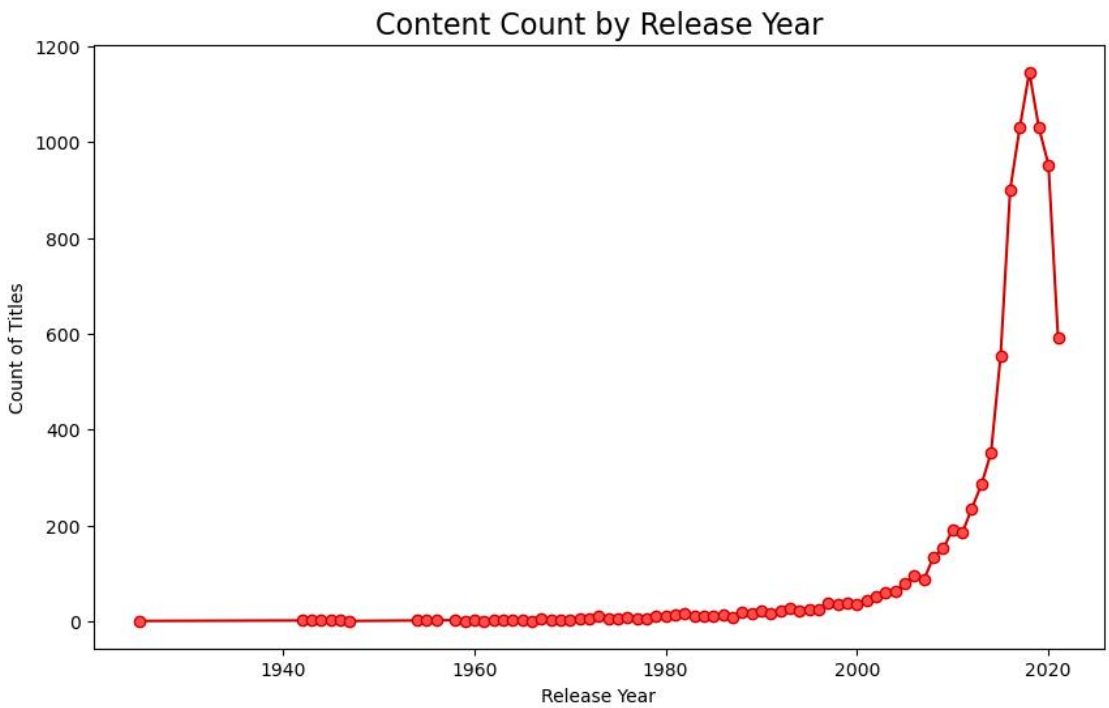
2) Content count by Release year

```
data['release_year'].value_counts()
```

```
2018    1146
2017    1030
2019    1030
2020     953
2016     901
      ...
1966       1
1959       1
1925       1
1947       1
1961       1
Name: release_year, Length: 74, dtype: int64
```

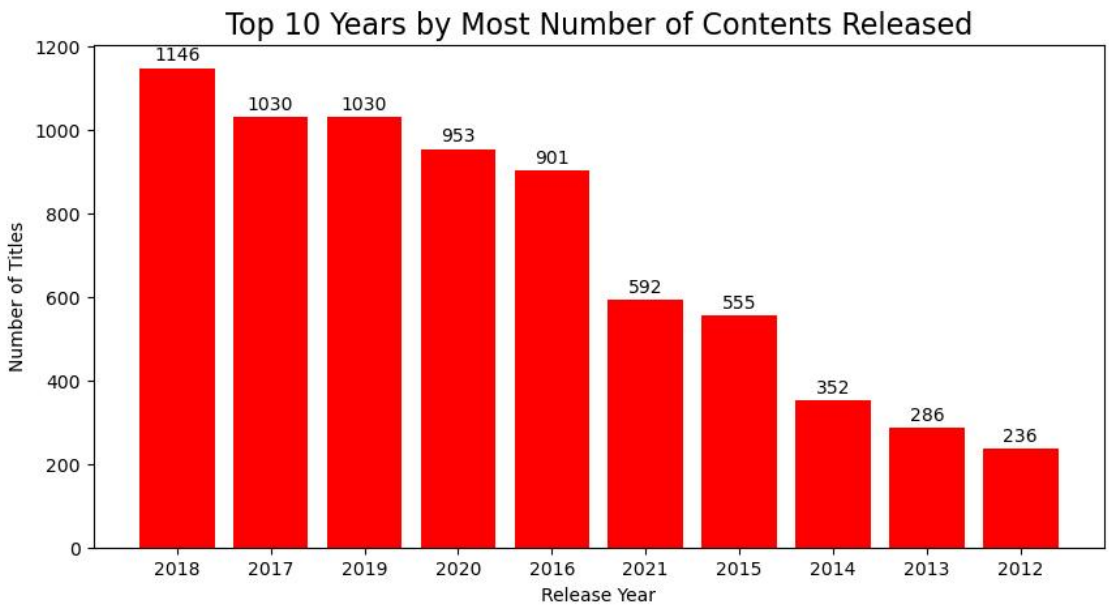
```
year_counts = data['release_year'].value_counts().sort_index()

plt.figure(figsize=(10,6))
plt.plot(year_counts.index,year_counts.values,marker='o',color='#cc0000',markerfacecolor='#ff4d4d')
```

```
plt.title('Content Count by Release Year', fontsize=16)
plt.xlabel('Release Year')
plt.ylabel('Count of Titles')
plt.grid(False)
plt.show()
```




```
# Calculate top 10 years by content released counts
top10_years = data['release_year'].value_counts().head(10).sort_values(ascending=Fals
e)

plt.figure(figsize=(10,5))

bars = plt.bar(top10_years.index.astype(str),top10_years.values,color='red')

plt.title("Top 10 Years by Most Number of Contents Released", fontsize=16)
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")
plt.grid(False)

for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + 10,
```

```
        str(height),
        ha='center',
        va='bottom',
        fontsize=10
    )

plt.show()
```



Findings:

Netflix started streaming in 2007. That event began the accumulation of releases.

Netflix content releases increased rapidly after 2015.

The highest number of titles were released in 2018.

2017 and 2019 were also strong years with over 1,000 releases each.

After 2019, the number of new releases started to decline.

The drop in 2020 and 2021 is likely due to COVID-19 production delays.

Very few titles exist before 1970, showing Netflix focuses mainly on modern and recent content.

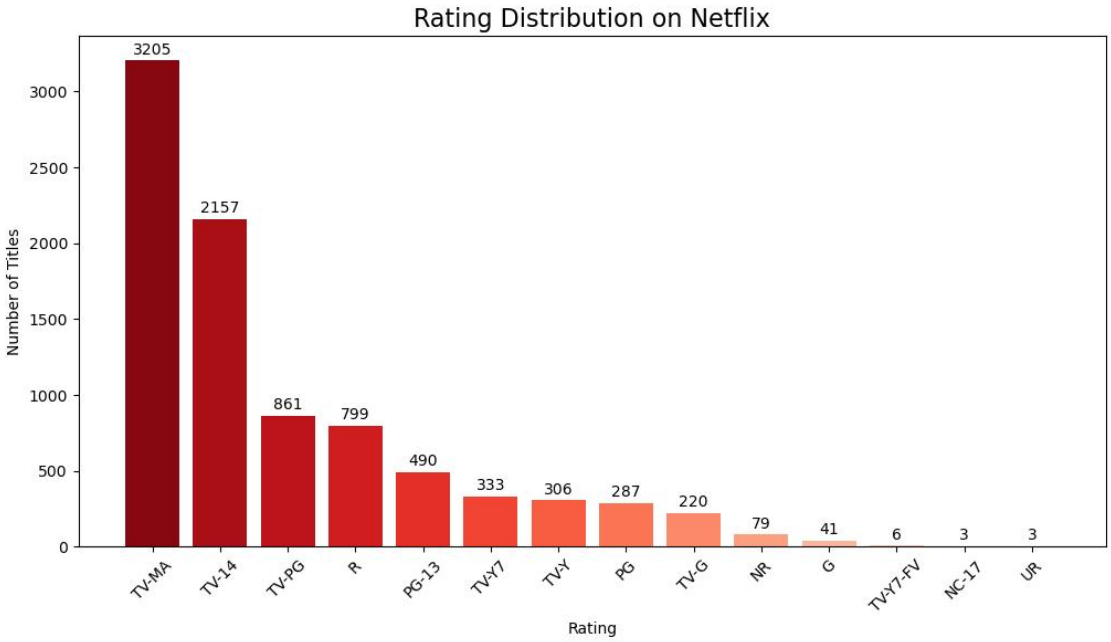
3) Rating Distribution of Content on Netflix

```
data['rating'].value_counts()
```

```
TV-MA      3205
TV-14      2157
```

```
TV-PG       861
R           799
PG-13       490
TV-Y7       333
TV-Y        306
PG          287
TV-G        220
NR           79
G            41
TV-Y7-FV      6
NC-17         3
UR            3
Name: rating, dtype: int64
```

```
plt.figure(figsize=(12,6))

rating_counts = data['rating'].value_counts()

bars = plt.bar(rating_counts.index,rating_counts.values,color=sns.color_palette("Reds_r
", len(rating_counts)))

plt.title("Rating Distribution on Netflix", fontsize=16)
plt.xlabel("Rating")
plt.ylabel("Number of Titles")
plt.grid(False)
plt.xticks(rotation=45)

for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + 20,
        str(height),
        ha='center',
        va='bottom',
        fontsize=10
    )

plt.show()
```



Findings:

TV-MA (Mature Audience) is the most common rating with 3,205 titles.

TV-14 is the second most frequent rating with 2,157 titles.

Family and kids-friendly ratings such as TV-PG, TV-Y7, TV-Y, and TV-G appear much less frequently.

Movie-specific adult ratings like R (799 titles) and PG-13 (490 titles) also form a significant portion of the library.

Very few titles are rated NC-17, UR, and TV-Y7-FV, showing minimal focus on extreme or niche rating categories.

Overall, Netflix content is heavily focused on teen and adult audiences rather than children.

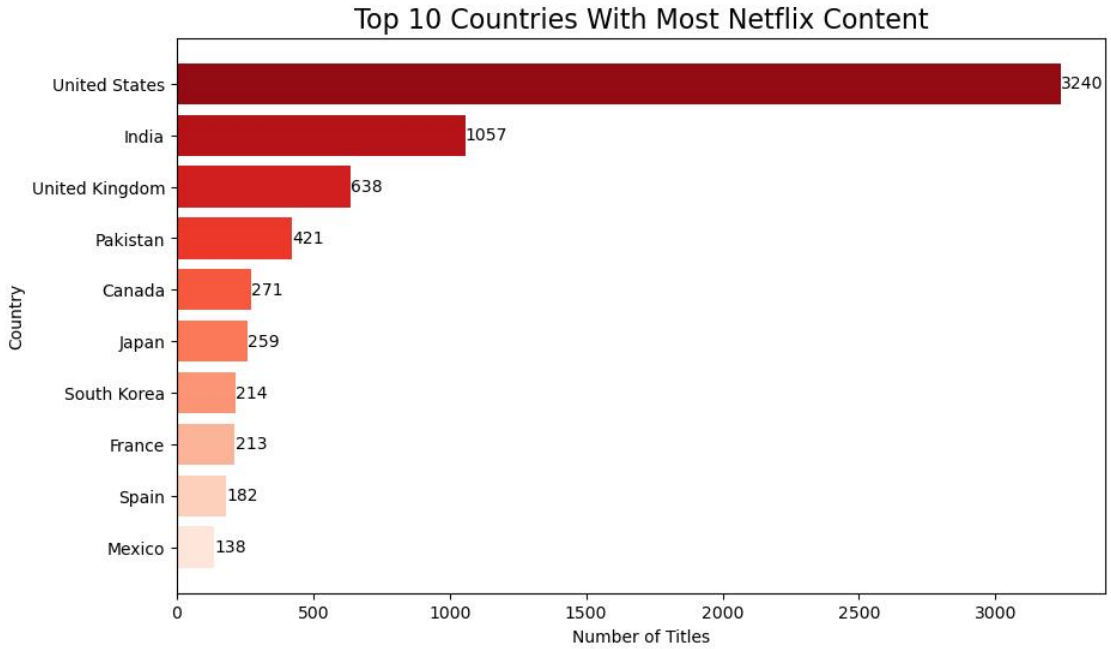
4) Countries With Most Content On Netflix

```
data['country'].value_counts()
```

```
United States     3240
India             1057
United Kingdom     638
Pakistan           421
Not Given          287
                 ...
Iran                 1
West Germany         1
```

```
Greece            1
Zimbabwe          1
Soviet Union      1
Name: country, Length: 86, dtype: int64
```

```
top10_countries = (
    data['country']
    .value_counts()
    .loc[lambda x: x.index != "Not Given"]
    .head(10)
    .reset_index()
    .rename(columns={'index': 'country', 'country': 'count'})
)

plt.figure(figsize=(10,6))

bars = plt.barh(top10_countries['country'],top10_countries['count'],color=sns.color_pal
ette("Reds_r", 10))

plt.title("Top 10 Countries With Most Netflix Content", fontsize=16)
plt.xlabel("Number of Titles")
plt.ylabel("Country")
plt.grid(False)

for bar in bars:
    width = bar.get_width()
    plt.text(
        width + 1,
        bar.get_y() + bar.get_height()/2,
        str(width),
        va='center',
        fontsize=10
    )

plt.gca().invert_yaxis()
plt.show()
```



Findings:

United States has the highest number of titles with 3,240 entries.

India ranks second with 1,057 titles, showing Netflix's strong focus on the Indian market.

United Kingdom is third with 638 titles.

Pakistan (421) and Canada (271) also contribute a significant amount of content.

Asian countries like Japan (259) and South Korea (214) appear in the top 10, reflecting the popularity of Asian content globally.

European countries such as France (213) and Spain (182) also have a strong presence.

Some entries have "Not Given" (287) as country, indicating missing or unrecorded country data.

Netflix's content library is dominated by the United States, but there is a rapidly growing contribution from India and other Asian countries, proving Netflix's global expansion strategy.

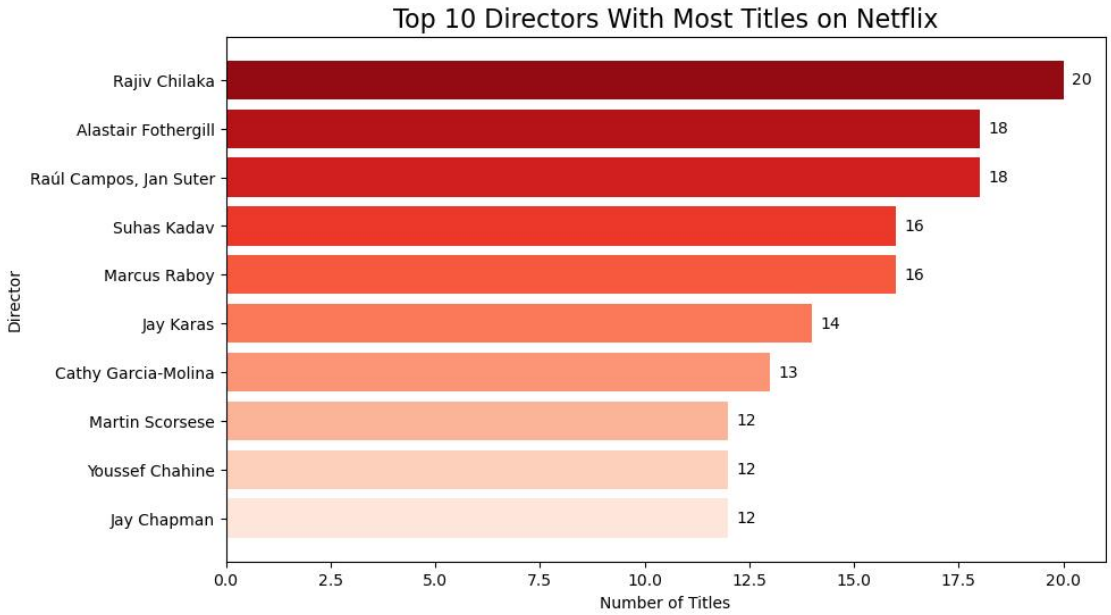
5) Directors with Most Titles

```
data['director'].value_counts()
```

```
Not Given                2588
Rajiv Chilaka              20
Alastair Fothergill        18
Raúl Campos, Jan Suter     18
```

```
Suhas Kadav                    16
                          ...
Matt D'Avella                   1
Parthiban                      1
Scott McAboy                    1
Raymie Muzquiz, Stu Livingston      1
Mozez Singh                     1
Name: director, Length: 4528, dtype: int64
```

```
top10_directors = (data['director'].value_counts().loc[lambda x: x.index != "Not Given"].
head(10))

plt.figure(figsize=(10,6))

bars = plt.barh(top10_directors.index,top10_directors.values,color=sns.color_palette("R
eds_r", 10))

plt.title("Top 10 Directors With Most Titles on Netflix", fontsize=16)
plt.xlabel("Number of Titles")
plt.ylabel("Director")
plt.grid(False)

for bar in bars:
    width = bar.get_width()
    plt.text(
        width + 0.2,
        bar.get_y() + bar.get_height()/2,
        str(width),
        va='center',
        ha='left',
        fontsize=10,
        color='black'
    )

plt.gca().invert_yaxis()
plt.show()
```



Findings:

Rajiv Chilaka leads with 20 titles.

Raúl Campos jJan Suter and Alastair Fothergill follow and each have 18 titles.

Marcus Raboy and Suhas Kadav have 16 titles each.

Jay Karas – 14 titles, Cathy Garcia-Molina – 13 titles, Martin Scorsese, Youssef Chahine, and Jay Chapman – 12 titles.

A large portion of entries (2,588) have "Not Given", showing missing director information for many titles.

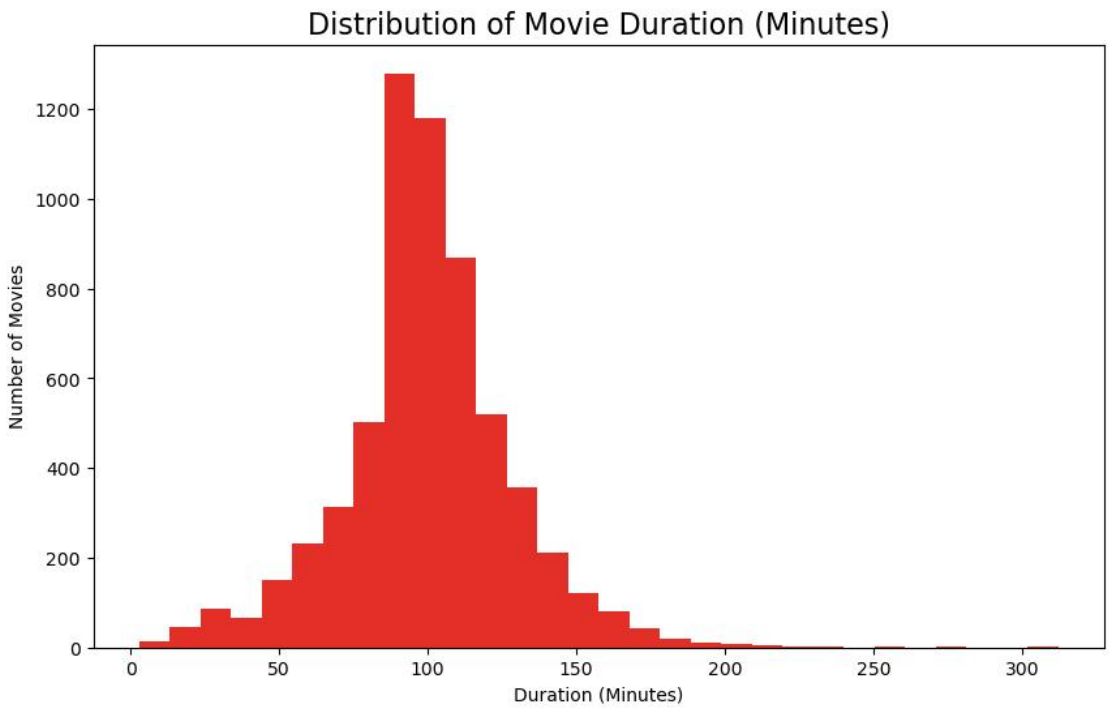
data.head(2)

```
  show_id     type              title        director        country  \
0      s1    Movie  Dick Johnson Is Dead  Kirsten Johnson  United States
1      s3  TV Show           Ganglands  Julien Leclercq         France

  date_added  release_year rating duration_text  \
0 2021-09-25          2020  PG-13       90 min
1 2021-09-24          2021  TV-MA     1 Season

                                listed_in  year_added  month_added  \
0                          Documentaries        2021            9
1  Crime TV Shows, International TV Shows, TV Act...        2021            9

   day_added  duration_int
```

0	25	90
1	24	1

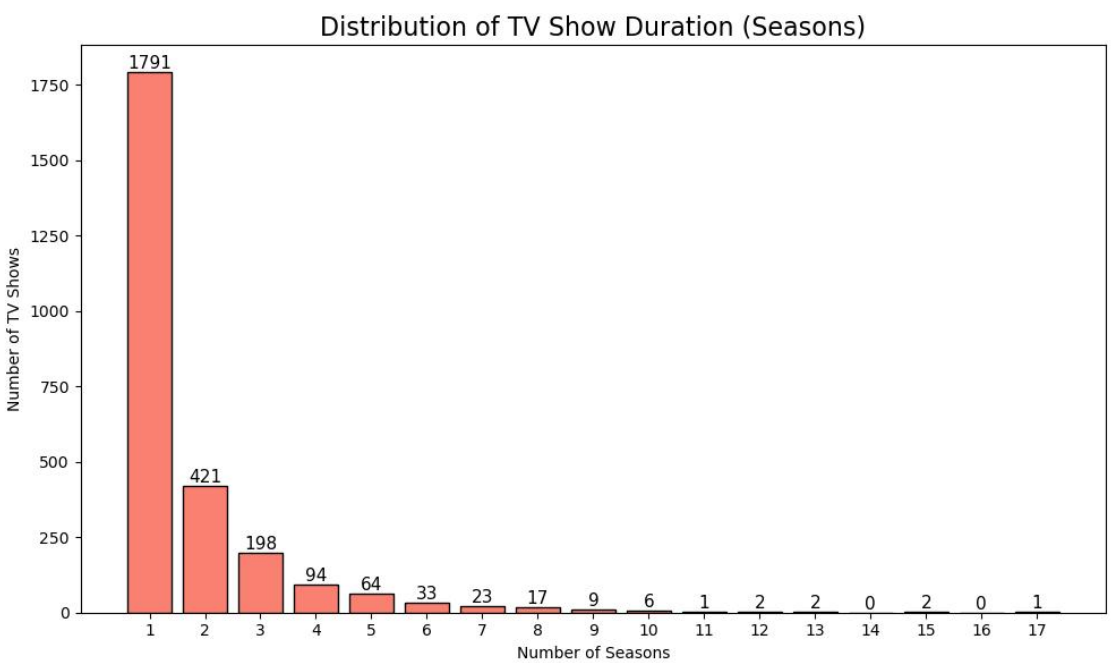
6) Duration Distribution of movies and TV shows

```
#Distribution of duration by movies

movies = data[data['type'] == 'Movie']
plt.figure(figsize=(10,6))
plt.hist(movies['duration_int'],bins=30,color=sns.color_palette("Reds", 5)[3])

plt.title("Distribution of Movie Duration (Minutes)", fontsize=16)
plt.xlabel("Duration (Minutes)")
plt.ylabel("Number of Movies")
plt.grid(False)

plt.show()
```




```
#Distribution of duration by TV shows

tv_shows = data[data['type'] == 'TV Show']
season_counts = tv_shows['duration_int'].value_counts().sort_index()
x_values = list(range(1, season_counts.index.max() + 1))
y_values = [season_counts.get(x, 0) for x in x_values]

plt.figure(figsize=(10,6))
```

```
bars = plt.bar(x_values,y_values,color='salmon',edgecolor='black')

plt.title("Distribution of TV Show Duration (Seasons)", fontsize=16)
plt.xlabel("Number of Seasons")
plt.ylabel("Number of TV Shows")
plt.grid(False)

for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + 0.1,
        str(int(height)),
        ha='center',
        va='bottom',
        fontsize=11
    )

plt.xticks(x_values)
plt.tight_layout()
plt.show()
```



Findings:

-Movies

Average movie length is around 100 minutes.

Most movies fall between 85 to 115 minutes.

The shortest movie is 3 minutes, and the longest is 312 minutes.

-TV Shows

Most TV shows have only 1 season.

Average number of seasons is about 1.7 seasons.

Very few shows run longer than 5 seasons, with the maximum reaching 17 seasons.

It shows that Netflix mainly focuses on standard-length movies and short-running TV series rather than long multi-season shows.

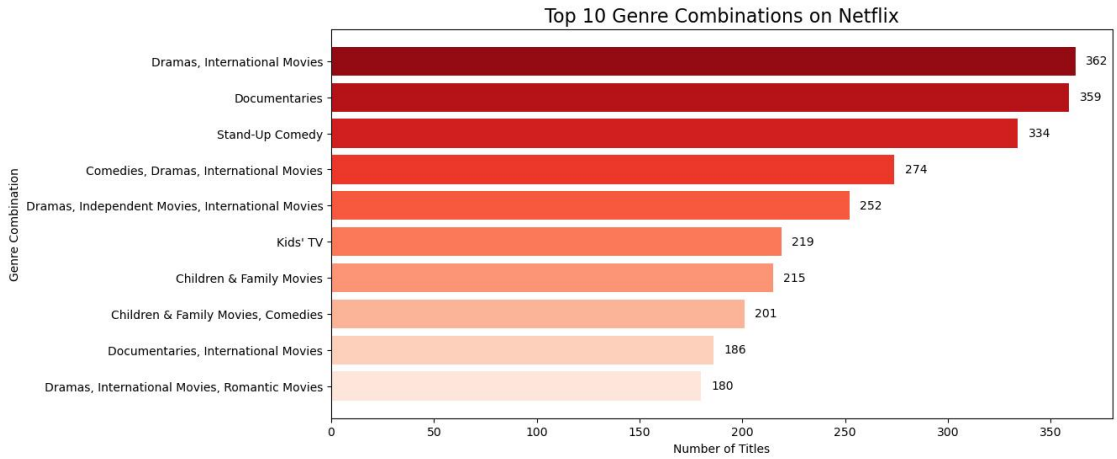
7) Genres with Most Contents

```
data['listed_in'].value_counts()
```

```
Dramas, International Movies                          362
Documentaries                                         359
Stand-Up Comedy                                       334
Comedies, Dramas, International Movies                274
Dramas, Independent Movies, International Movies      252
                                                      ...
Anime Features                                        1
Action & Adventure, Horror Movies, Independent Movies   1
Action & Adventure, Classic Movies, International Movies   1
Cult Movies, Independent Movies, Thrillers            1
Classic & Cult TV, Crime TV Shows, TV Dramas          1
Name: listed_in, Length: 513, dtype: int64
```

```
top10_genres_raw = data['listed_in'].value_counts().head(10)
plt.figure(figsize=(12,6))

bars = plt.barh(top10_genres_raw.index[::-1],top10_genres_raw.values[::-1],color=sns.c
olor_palette("Reds", 10))

plt.title("Top 10 Genre Combinations on Netflix", fontsize=16)
plt.xlabel("Number of Titles")
plt.ylabel("Genre Combination")
plt.grid(False)

for bar in bars:
    width = bar.get_width()
    plt.text(
```

```
        width + 5,
        bar.get_y() + bar.get_height()/2,
        str(width),
        va='center',
        fontsize=10
    )

plt.show()
```



Findings:

“Dramas, International Movies” is the most common genre combination with 362 titles.

Documentaries alone appear 359 times, showing strong demand for non-fiction content.

Stand-Up Comedy ranks third with 334 titles, highlighting Netflix’s focus on comedy specials.

“Comedies, Dramas, International Movies” is another popular combination with 274 titles.

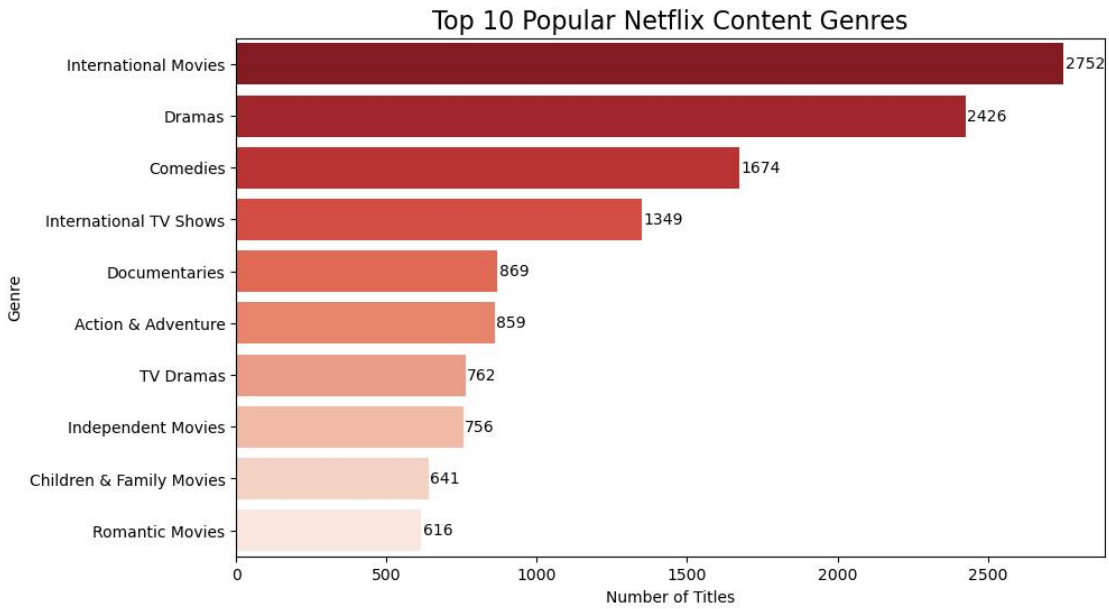
“Dramas, Independent Movies, International Movies” appears 252 times, showing Netflix’s emphasis on global and indie drama content.

Netflix has over 500 unique genre combinations, proving the platform offers a highly diverse content mix.

Many genre combinations occur only once, indicating experimentation with niche and hybrid content styles.

Netflix’s most frequent genre combinations revolve around Dramas, International content, Documentaries, and Stand-Up Comedy, reflecting its strategy to deliver emotionally engaging, globally relevant, and variety-driven entertainment.

```
top_genres = genres_data['genre'].value_counts().head(10)

plt.figure(figsize=(10,6))
sns.barplot(x=top_genres.values, y=top_genres.index, palette='Reds_r')

for index, value in enumerate(top_genres.values):
    plt.text(value + 5, index, str(value), va='center')

plt.title('Top 10 Popular Netflix Content Genres', fontsize=16)
plt.xlabel('Number of Titles')
plt.ylabel('Genre')
plt.show()
```



Findings:

Netflix strongly focuses on International content and Dramas, showing its strategy to target a global audience with emotionally engaging stories, while also maintaining a healthy mix of comedies, documentaries, and family-oriented genres.

8) Monthly and Yearly Additions of Contents on Netflix

```
monthly_movie_additions = data[data['type'] == 'Movie']['month_added'].value_counts
().sort_index()
monthly_movie_additions
```

1 545
2 382
3 528

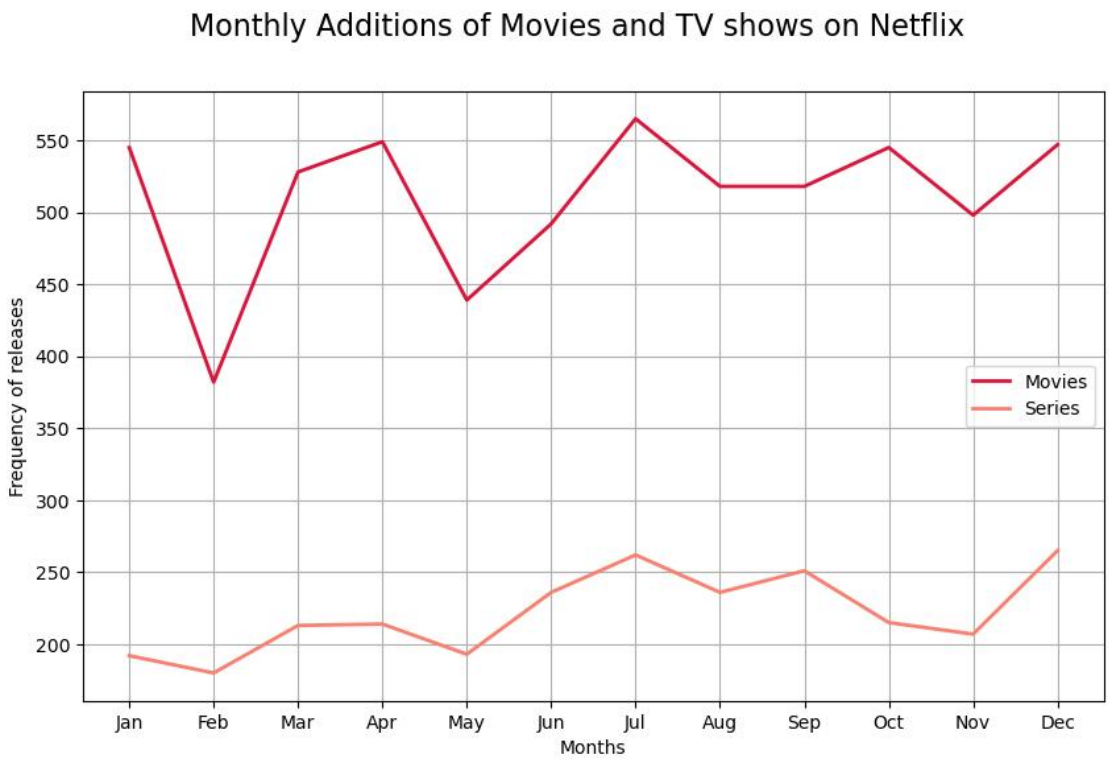
```
4    549
5    439
6    492
7    565
8    518
9    518
10   545
11   498
12   547
Name: month_added, dtype: int64
```

```
monthly_series_additions = data[data['type'] == 'TV Show']['month_added'].value_coun
ts().sort_index()
monthly_series_additions
```

```
1    192
2    180
3    213
4    214
5    193
6    236
7    262
8    236
9    251
10   215
11   207
12   265
Name: month_added, dtype: int64
```

```
monthly_movie_additions = data[data['type'] == 'Movie']['month_added'].value_counts
().sort_index()
monthly_series_additions = data[data['type'] == 'TV Show']['month_added'].value_coun
ts().sort_index()
plt.figure(figsize=(10,6))
plt.plot(monthly_movie_additions.index, monthly_movie_additions.values, label='Movi
es', color='crimson', linewidth=2)
plt.plot(monthly_series_additions.index, monthly_series_additions.values, label='Series',
 color='salmon', linewidth=2)
plt.xlabel("Months")
plt.ylabel("Frequency of releases")
plt.xticks(range(1, 13), ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov',
'Dec'])
plt.legend()
plt.grid(True)
```

```
plt.suptitle("Monthly Additions of Movies and TV shows on Netflix", fontsize=16)
plt.show()
```



Findings:

Netflix adds the highest number of titles in July and December.

February has the lowest number of releases for both Movies and TV Shows.

Movie additions remain fairly consistent throughout the year.

TV Show additions show more fluctuation compared to Movies.

The second half of the year (July–December) records more content additions than the first half.

Peak months for Movies: July, December, April, January, and October.

Peak months for TV Shows: December, July, September, June, and August.

Netflix likely schedules more releases during holiday and festive seasons to increase viewer engagement.

These patterns clearly show Netflix's strategy of releasing more content in the second half of the year and maintaining a stronger focus on movie releases compared to TV series.

```
yearly_movie_added = data[data['type'] == 'Movie']['year_added'].value_counts().sort_index()
yearly_movie_added
```

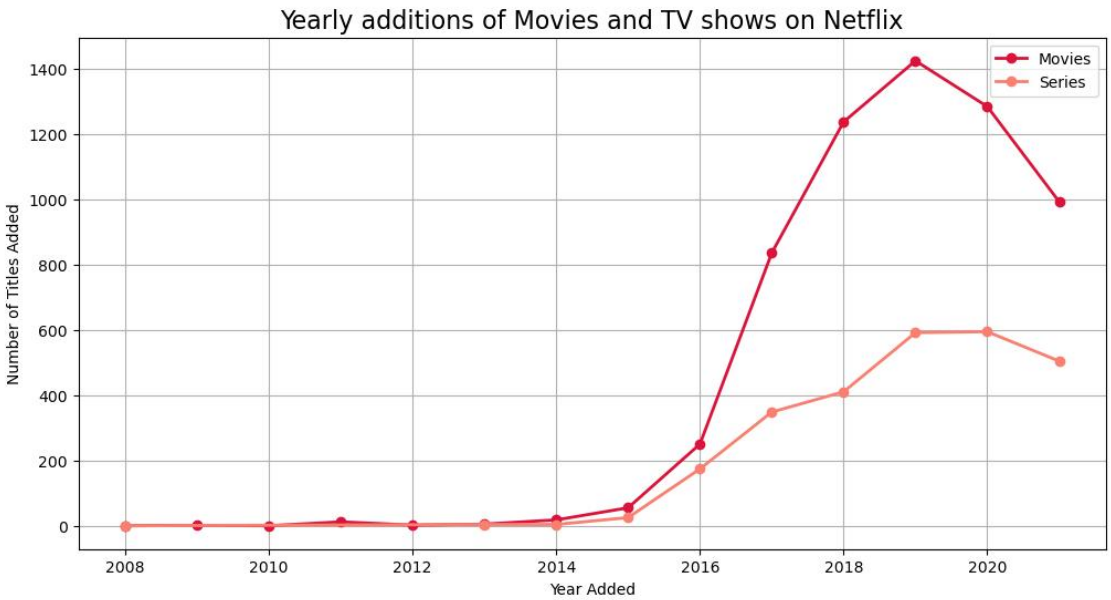
```
2008      1
2009      2
2010      1
2011     13
2012      3
2013      6
2014     19
2015     56
2016    251
2017    836
2018   1237
2019   1424
2020   1284
2021    993
Name: year_added, dtype: int64
```

```
yearly_series_added = data[data['type'] == 'TV Show']['year_added'].value_counts().sort_index()
yearly_series_added
```

```
2008      1
2013      5
2014      5
2015     26
2016    175
2017    349
2018    411
2019    592
2020    595
2021    505
Name: year_added, dtype: int64
```

```
yearly_movie_added = data[data['type'] == 'Movie']['year_added'].value_counts().sort_index()
yearly_series_added = data[data['type'] == 'TV Show']['year_added'].value_counts().sort_index()
plt.figure(figsize=(12,6))
plt.plot(yearly_movie_added.index, yearly_movie_added.values, label='Movies', color='crimson', linewidth=2, marker='o')
plt.plot(yearly_series_added.index, yearly_series_added.values, label='Series', color='salmon', linewidth=2, marker='o')
```

```
plt.xlabel("Year Added")
plt.ylabel("Number of Titles Added")
plt.title("Yearly additions of Movies and TV shows on Netflix", fontsize=16)
plt.legend()
plt.grid(True)
plt.show()
```



Findings:

Netflix added very little content before 2015.

Netflix content additions increased steadily from 2015 onwards.

Movie additions peaked in 2019.

TV Show additions peaked in 2020.

The highest growth is seen between 2017 and 2019 for both Movies and TV Shows.

After 2019, there is a visible decline in new additions.

The reduction in 2020 and later years suggests pandemic-related production slowdowns.

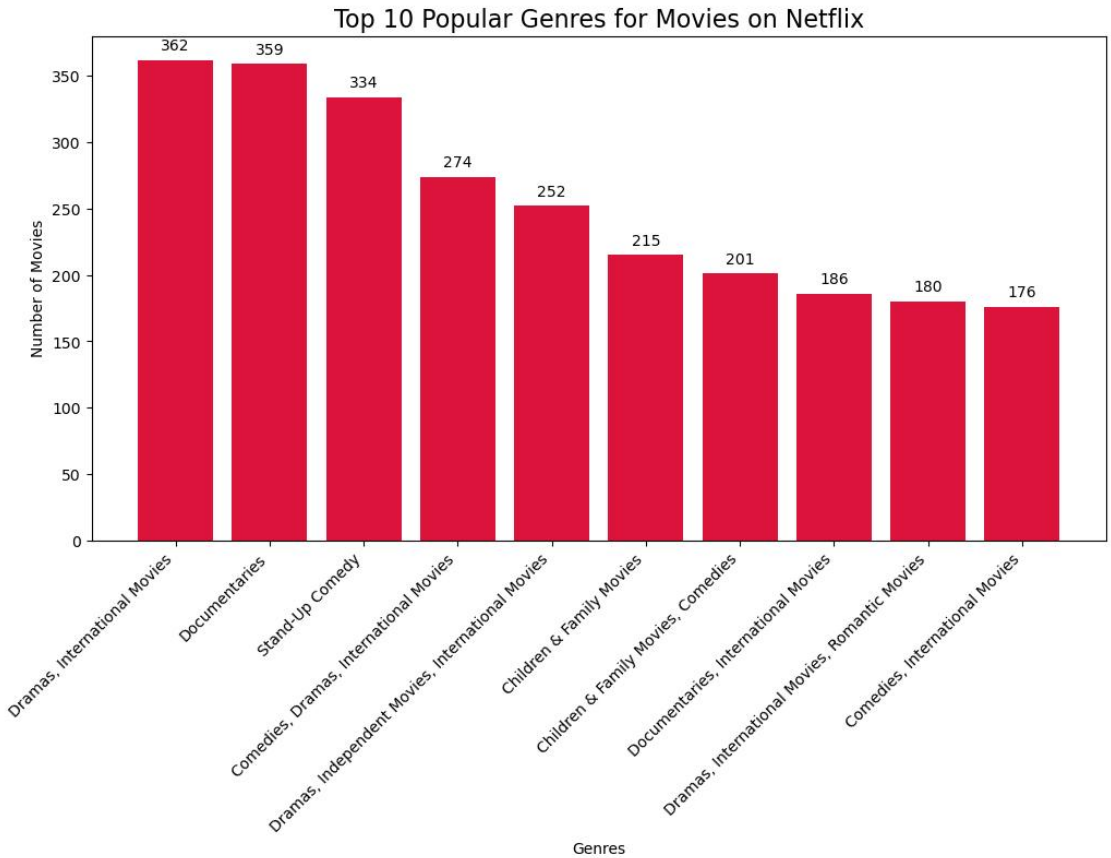
Throughout all years, Movies consistently outnumber TV Shows in yearly additions.

The period 2016–2020 represents Netflix's major expansion phase.

9) Top Popular Genres for Movies and TV Shows on Netflix

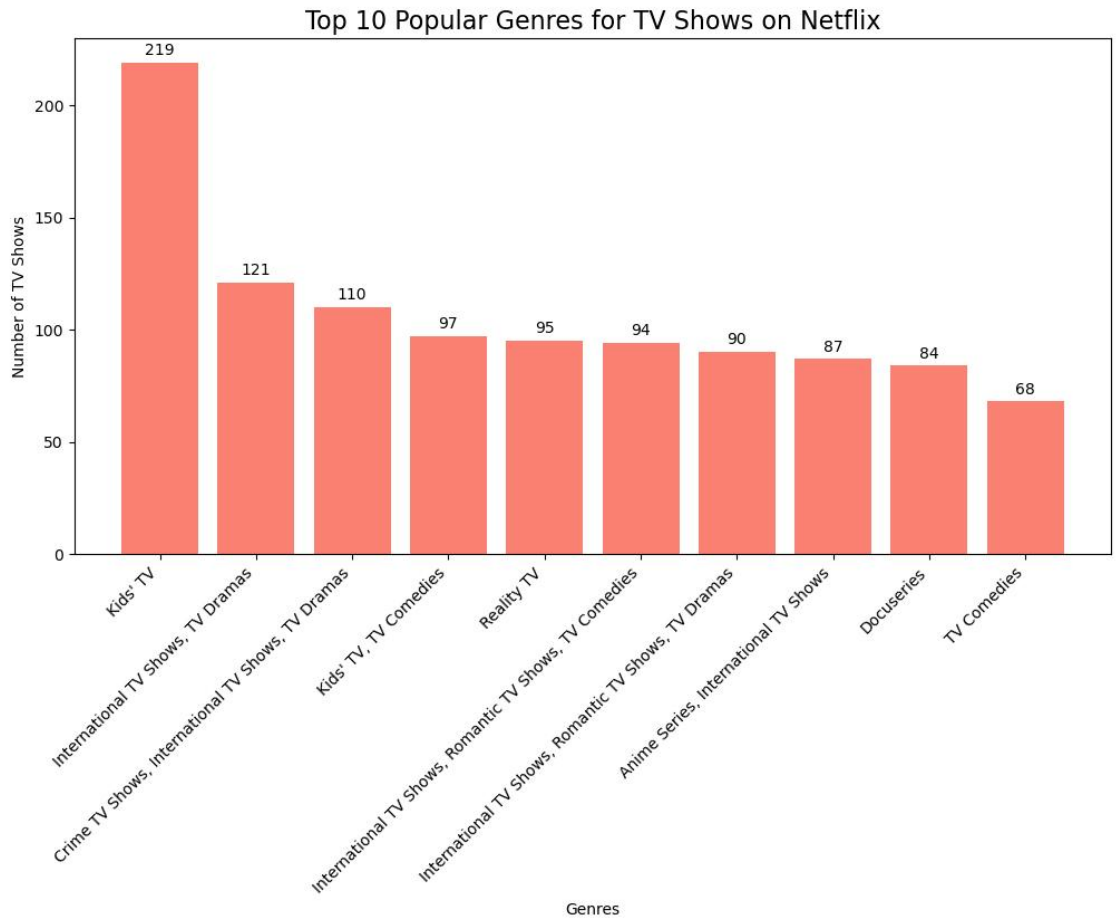
```
popular_movie_genre = data[data['type'] == 'Movie'].groupby("listed_in").size().sort_values(ascending=False)[:10]
plt.figure(figsize=(12,6))
```

```
bars = plt.bar(popular_movie_genre.index, popular_movie_genre.values, color='crimson
')
for bar in bars:
   height = bar.get_height()
   plt.text(bar.get_x() + bar.get_width()/2, height + 5, str(height), ha='center', va='botto
m')

plt.xticks(rotation=45, ha='right')
plt.xlabel("Genres")
plt.ylabel("Number of Movies")
plt.title("Top 10 Popular Genres for Movies on Netflix", fontsize=16)
plt.show()
```




```
popular_series_genre = data[data['type'] == 'TV Show'].groupby("listed_in").size().sort_v
alues(ascending=False)[:10]
plt.figure(figsize=(12,6))
bars = plt.bar(popular_series_genre.index, popular_series_genre.values, color='salmon')
for bar in bars:
   height = bar.get_height()
   plt.text(bar.get_x() + bar.get_width()/2, height + 2, str(height), ha='center', va='botto
m')
```

```
plt.xticks(rotation=45, ha='right')
plt.xlabel("Genres")
plt.ylabel("Number of TV Shows")
plt.title("Top 10 Popular Genres for TV Shows on Netflix", fontsize=16)
plt.show()
```



Findings:

-Movies

Dramas dominate the movie library, often combined with International or Independent Movies.

Documentaries are highly represented, highlighting Netflix's focus on educational and factual content.

Stand-Up Comedy ranks third, showing strong popularity of comedy specials.

Family-friendly content appears consistently, with multiple Children & Family Movies in top categories.

Many movies are multi-genre, often combining Drama, Comedy, International, or Romantic categories.

Netflix's movie library is heavily skewed towards dramas, documentaries, and international content, with comedy and family movies also well represented.

-TV Shows

Kids' TV shows are the most common, indicating Netflix's focus on family and children's content.

International TV Shows and Dramas are popular, reflecting global audience appeal.

Crime, Reality, and Anime appear frequently, showing demand for diverse entertainment.

Multi-genre shows are common, combining elements like Drama, Comedy, and International themes.

Netflix's TV library focuses on kids and international dramas, with variety across reality, crime, and anime, appealing to a broad global audience.

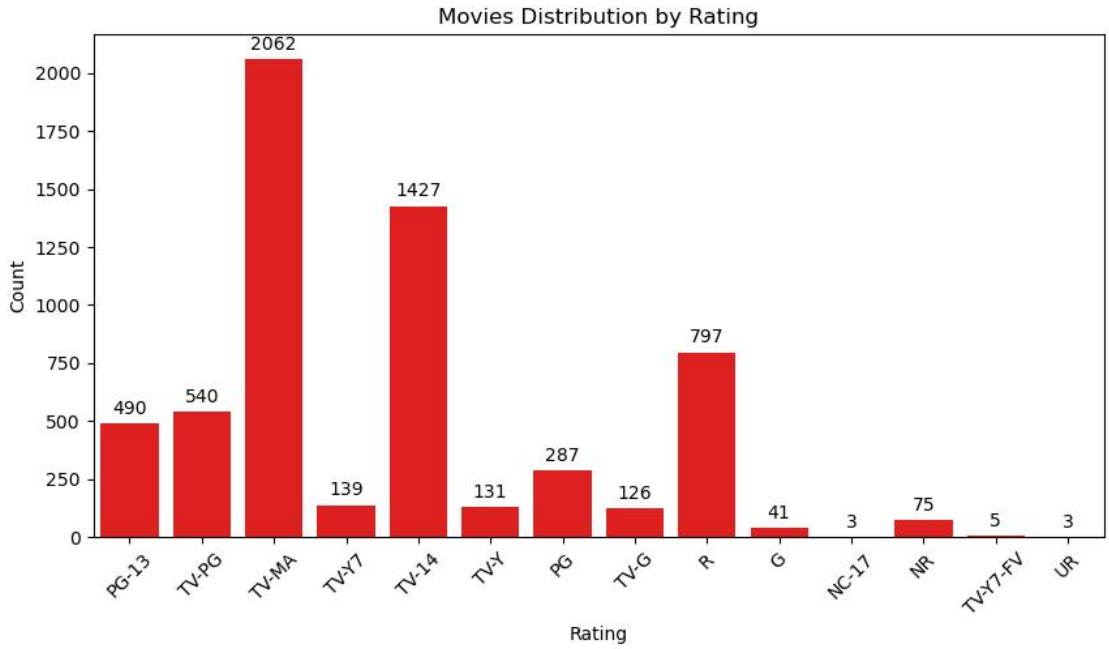
Netflix prioritizes a balance between adult dramas, educational documentaries, and family/kids content.

Multi-genre content is a key strategy to cater to diverse viewer preferences.

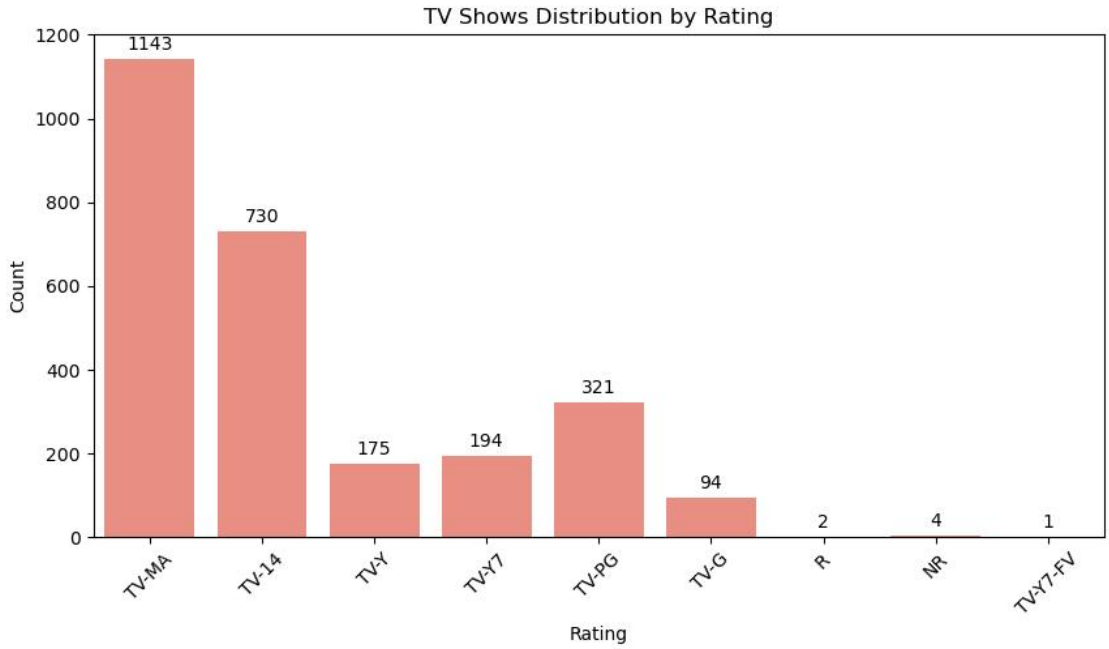
Popular genres reflect global appeal, cross-generational content, and a mix of entertainment and educational offerings.

10) Count of Movies and TV Shows by Rating

```
plt.figure(figsize=(10,5))
movie_data = data[data['type'] == 'Movie']
ax = sns.countplot(x='rating', data=movie_data, color='red')
plt.xticks(rotation=45)
plt.xlabel('Rating')
plt.ylabel('Count')
plt.title('Movies Distribution by Rating')
for p in ax.patches:
    height = int(p.get_height())
    ax.annotate(f'{height}',
                (p.get_x() + p.get_width()/2., height),
                ha='center', va='bottom', fontsize=10, xytext=(0,3),
                textcoords='offset points')

plt.show()
```




```
plt.figure(figsize=(10,5))
series_data = data[data['type'] == 'TV Show']
ax = sns.countplot(x='rating', data=series_data, color='salmon')
plt.xticks(rotation=45)
plt.xlabel('Rating')
plt.ylabel('Count')
plt.title('TV Shows Distribution by Rating')
for p in ax.patches:
    height = int(p.get_height())
    ax.annotate(f'{height}',
                (p.get_x() + p.get_width()/2., height),
                ha='center', va='bottom', fontsize=10, xytext=(0,3),
                textcoords='offset points')

plt.show()
```



Findings:

-Movies

TV-MA (Mature audiences) is the most common rating, followed by TV-14 and R, indicating a focus on adult and teen viewers.

Family/kids content (TV-Y, TV-Y7, TV-G, G) is limited.

Restricted content (NC-17, TV-Y7-FV) is very rare.

-TV Shows

TV-MA shows dominate, targeting mature audiences.

TV-14 is next, appealing to teenagers.

TV-PG provides moderately family-friendly content for general audiences with parental guidance.

Kids content (TV-Y, TV-Y7, TV-G) exists but forms a smaller segment.

Restricted content (R, NR, TV-Y7-FV) is minimal.

Netflix's library skews heavily toward adult and teen audiences, with TV-PG bridging moderate content for family/teen viewers.

Kids/family content is present but smaller in proportion.

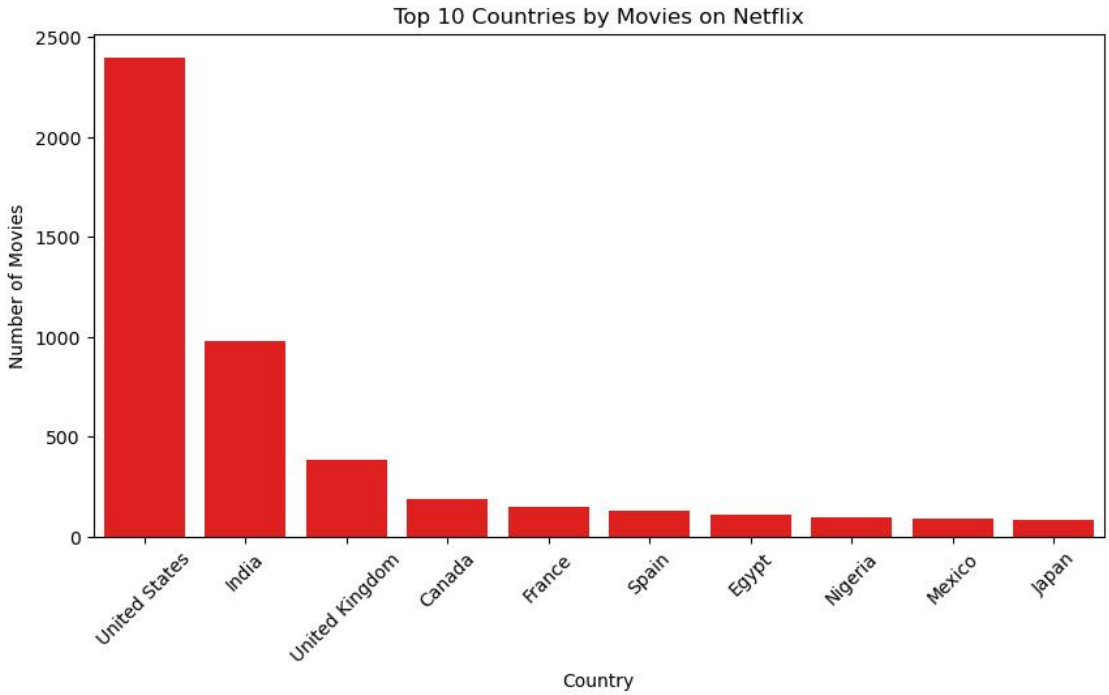
The distribution aligns with Netflix's strategy of offering diverse adult entertainment while keeping highly restricted content limited.

11) Top Countries producing Movies and TV Shows

```
top_countries_movies = (data[(data['type'] == "Movie") &(data['country'] != "Not Given
")]['country'].value_counts().head(10))
print(top_countries_movies)
```

```
United States     2395
India             976
United Kingdom    387
Canada            187
France            148
Spain             129
Egypt             109
Nigeria           96
Mexico            90
Japan             87
Name: country, dtype: int64
```

```
top_countries_movies = top_countries_movies.reset_index()
top_countries_movies.columns = ['country', 'count']

plt.figure(figsize=(10,5))
sns.barplot(data=top_countries_movies, x='country', y='count', color='red')

plt.xticks(rotation=45)
plt.xlabel('Country')
plt.ylabel('Number of Movies')
plt.title('Top 10 Countries by Movies on Netflix')
plt.show()
```

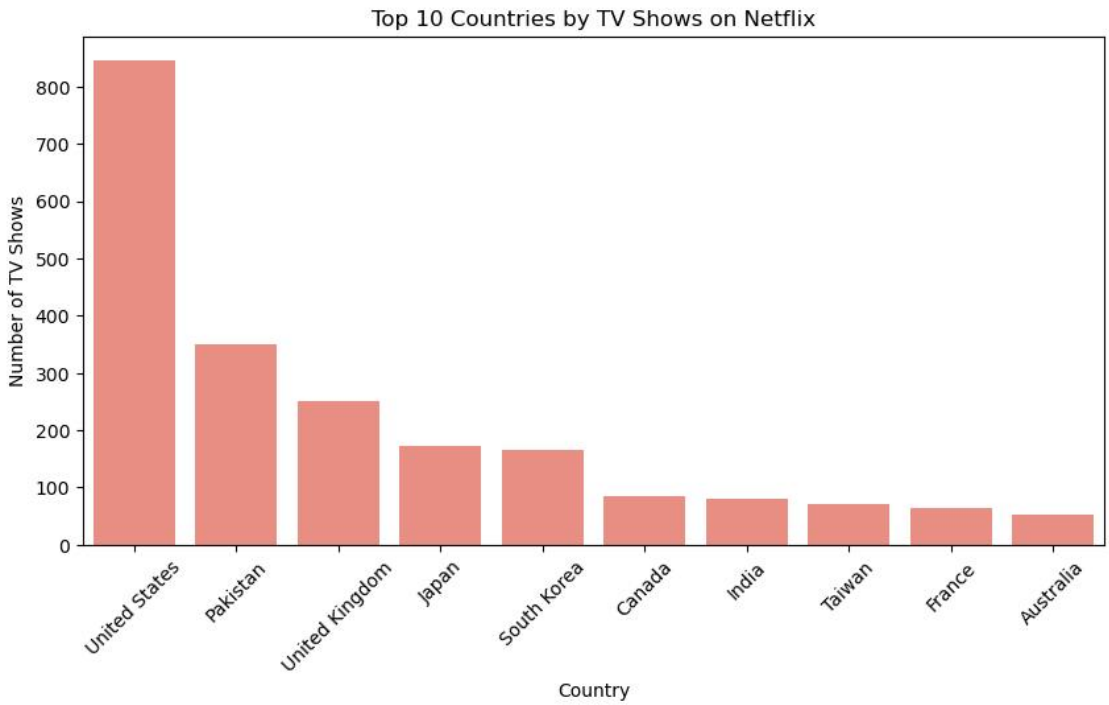



```
top_countries_tvshows = (data[(data['type'] == "TV Show") &(data['country'] != "Not Giv
en")]['country'].value_counts().head(10))
print(top_countries_tvshows)
```

```
United States     845
Pakistan          350
United Kingdom    251
Japan             172
South Korea       165
Canada             84
India              81
Taiwan             71
France             65
Australia          53
Name: country, dtype: int64
```

```
top_countries_tvshows = top_countries_tvshows.reset_index()
top_countries_tvshows.columns = ['country', 'count']

plt.figure(figsize=(10,5))
sns.barplot(data=top_countries_tvshows, x='country', y='count', color='salmon')

plt.xticks(rotation=45)
plt.xlabel('Country')
plt.ylabel('Number of TV Shows')
```

```
plt.title('Top 10 Countries by TV Shows on Netflix')
plt.show()
```



Findings:

US Dominance: The United States leads Netflix's library, providing the most content for both Movies (2,395 titles) and TV Shows (845 titles).

India's Movie Power: India is the second-largest source of movies, largely due to Bollywood (976 titles), but contributes significantly fewer TV shows (81 titles).

Asian TV Hubs: Japan and South Korea are key contributors to TV shows, driven by the global popularity of Anime and K-Dramas.

Regional Movie Content: Countries like Nigeria and Egypt have strong movie representation, showing Netflix's success in capturing regional film industries.

Concentrated TV Production: TV shows are more concentrated in specific markets such as Pakistan and the UK, whereas movie production is geographically more diverse.

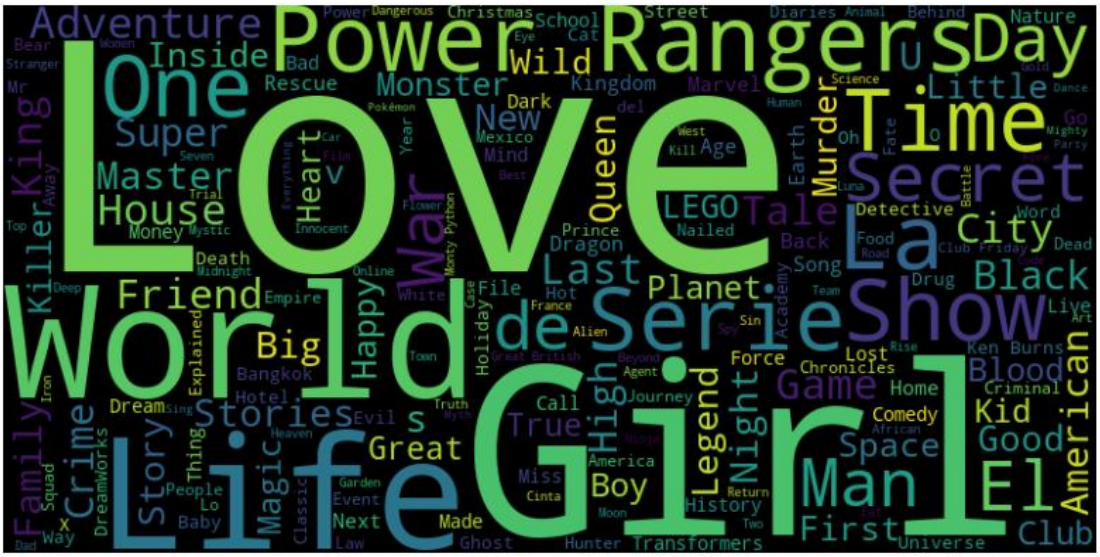
Netflix's content library reflects a US-heavy focus, with strategic investments in regional movie markets and international TV hubs, ensuring both global and local audience appeal.

WordCloud for Movies, and TV Shows

```
# Generate word cloud for Movie titles
movie_titles = data[data['type'] == 'Movie']['title']
```

```
wordcloud = WordCloud(width=800, height=400,
background_color='black').generate(' '.join(movie_titles))
plt.figure(figsize=(10, 6))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.show()
```

Movie
Love
Christmas
Story
Black
Last
Girl
Secret
Day
Night
Boy
King
Little
House
Dream
City
Man
World
Life
One
Time
Live
American
War
Special
Game
Princess
Shadow
Welcome
Happy
Death
Dragon
Woman
Sister
Lost
Back
Thing
Baby
Adventure
Battle
Ghost
Return
Take
Show
Money
Blood
Red
Star
Road
Ki
Murder
God
El
Hot
Holiday
John
Good
Prince
Master
Family
Island
London
Party
Earth
Bad
Dark
Chhota
Bheem
Great
Dead
Never
Monster
Devil
Mission
Wedding
First
Rise
Legend
Sky
Tale
Barbie
Brother
Cop
Ka
Mr
Friend
Right
Sea
Fire
Present
Best
Bird
Born
Big
Eye
Song
Dil
Two
Blue
II
Mother
Afterparty
Monster High
Kung Fu
Journey
Motu Patlu
Space
Everything
Let
Wild
U
Heart
Home
Free
Halloween
America
Part
Planet
Crime
Super Monster
Year
Alien
State
Beginning
Son
Seven
Stand
Miss
Wish
Tiger
Animal
Rock
Shot
Beyond
Escape
del
Breaking
People
Magic
Oh
Old
Stories
End
La
Dance
History
High
Room
True
Father
Perfect
Fall
Kill
New
Now
Forever
Wanted
Kid
Light
NOVA
Lady
Men
Aur
Hai
Real
Water
Hour
Moon
Mumbai
Land
Boss
Know
Dog
Ghost
Mother
Next
Blue
Film
Women
Born
Best
Present
Great
Bird
Bheem
Chhota
Dark
Earth
Party
Never
Dead
Monster
Devil
Mission
Wedding
First
Rise
Legend
Sky
Tale
Barbie
Brother
Cop
Ka
Mr
Friend
Right
Sea
Fire
Present
Best
Bird
Great
Bheem
Chhota
Dark
Bad
Earth
Party
London
Island
Two
Special
Miss
Wish
Tiger
Animal
Rise


Love
World
Life
Girl
Power
Rangers
Day
Time
Secret
La
City
Black
Show
Serie
de
Man
El
American
Adventure
One
Inside
Super
Master
House
King
Killer
Friend
Big
Stories
Story
Crime
Family
Magic
Great
Next
Made
Boy
True
Last
Planet
Game
Space
First
Club
Good
Kid
Blood
Night
Legend
High
Queen
War
Heart
Monster
Wild
New
Rescue
Little
Murder
Tale
LEGO
Dragon
Prince
Dark
Kingdom
Marvel
Mexico
Mind
Money
Death
Empire
Happy
Holiday
File
Hot
Alien
Force
Lost
Chronicles
Home
Comedy
Criminal
Ken Burns
Detective
Food
Drug
Song
Back
Nailed
Age
Earth
Oh
Bangkok
Hotel
Dream
Call
Journey
Miss
Cinta
America
History
Return
Transformers
Universe
Hunter
Ghost
Garden
Event
Baby
People
Lo
Thing
Explained
Dreamworks
Squad
Way
Bad
Law
Classic
Evil
Truth
Heaven
Sing
Iron
Toxin
Great British
Beyond
Agent
Sin
Spy
Rise
Art
Live
Burns
Criminal
Club Friday
Dead
Road
Battle
Luna
Team
Academy
Drama
Flower
Python
Monty
Case
Online
White
Innocent
Trial
Mystic
Seven
Everything
Car
Film
Year
Pokémon
Human
Science
Dance
Mighty
Party
Go
Fate
Oh
Kill
West
Marvel
Cat
Eye
Bad
Stranger
Bear
Mr
Away
Top
Deep
Dangerous
Power
Christmas
School
Street
Diaries
Animal
Behind
Nature
Gold
Dance
Human
Science
Word
Dead
Live
Art
Rise
Ken Burns
Criminal
Club Friday
Road
Food
Drug
Team
Academy
Sin
Agent
Journey
Return
Two
Moon
Cinta
Universe
Hunter
Transformers
Law
Garden
Event
Garden
Classic
Dreamworks
Squad
Way
Bad
Iron
Sing
Heaven
Truth
Evil
Hotel
Bangkok
Explained
Thing
Lo
Baby
People
Dream
Crime
Story
Stories

TV-MA is the top rating, followed by TV-14.

TV-PG (321 titles) acts as a bridge category which means moderate family-friendly content.

Kids-only ratings form a minority segment.

Netflix's catalogue is clearly skewed toward mature content, not family-centric entertainment.

3. Genre Trends

Top Movie Genres are dominated by Dramas, International Movies, Documentaries and Stand-Up Comedy

Heavy presence of multi-genre combinations (Drama + International, Drama + Romance, Comedy + International).

Top TV Show Genres are led by Kids' TV, International TV Shows, TV Dramas, Crime TV Shows, International TV Shows, TV Dramas. Anime & Docuseries also appear frequently.

Netflix's movie strategy = Drama + Documentary + Comedy.

Netflix's TV strategy = Kids + International Dramas + Crime + Anime.

4. Country-wise Content Distribution

- Movies

USA is rank #1 by a huge margin.

India is rank #2 (Bollywood dominance).

Nigeria & Egypt have strong presence which means Netflix has tapped Nollywood & Middle-Eastern cinema.

- TV Shows

USA leads, but Pakistan ranks #2.

Japan & South Korea are strong contributors (Anime & K-Drama effect).

India contributes far fewer TV shows than movies.

Movies are globally diversified.

TV shows are geographically concentrated in a few production hubs.

5. Growth Over Time

Sharp rise in Netflix content after 2015.

Major spike between 2017–2020 which shows aggressive global expansion phase.

6. Missing & Incomplete Data

Large number of records with missing director names and multiple genres in a single field indicates inconsistent metadata standards across regions.

Strategic Recommendations:

1. Content Strategy

- Expand Family & Kids Content
- Current library is overly adult-oriented.
- Increasing kids/family movies will improve household subscriptions.
- Invest More in Indian & Asian TV Shows
- India dominates movies but underperforms in TV shows.
- Replicate South Korea & Japan's TV model in India.
- Double Down on International Dramas
- International TV Dramas consistently rank among top genres.
- Localized storytelling is Netflix's biggest competitive edge.
- Boost Documentary + Stand-Up Production
- These genres perform strongly in movies and require lower production cost with high ROI.

2. Market Expansion

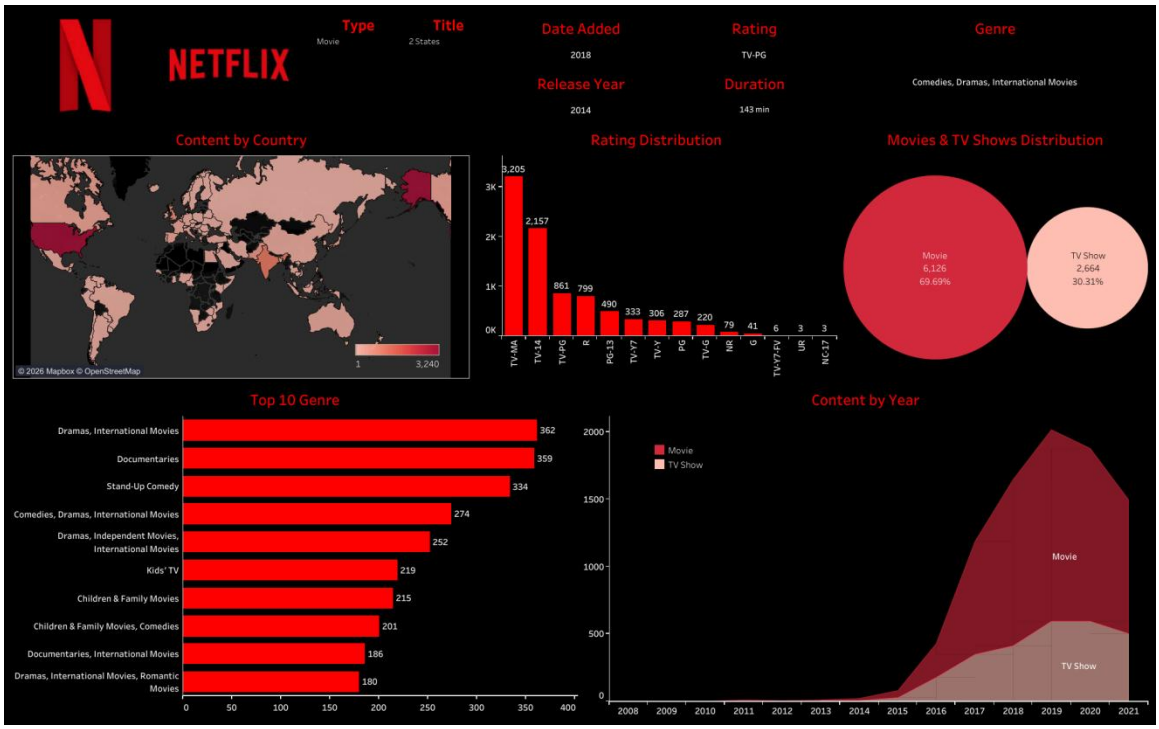
- Strengthen Regional Film Industries
- Nigeria (Nollywood), Egypt, Mexico show strong movie output.
- Create Netflix Originals in these regions to dominate untapped markets.
- Reduce Over-reliance on the US
- US content dominance exposes Netflix to saturation risk.
- Balanced global sourcing will future-proof the library.

3. Data & Operations

- Standardize Genre Metadata
- Split multi-genre values into separate records for accurate analytics.

- Improve Missing Director Attribution
- Missing values reduce search relevance & recommendation accuracy.
- Content Lifecycle Monitoring
- Track performance by release year to identify genres losing relevance and markets needing reinvestment

Conclusion:

This project analyzed the Netflix content catalog using data cleaning,feature engineering, and exploratory data analysis to uncover key business insights. The findings show that N etflix's library is movie-dominant, primarily targets mature and teen audiences, and stro ngly focuses on dramas and international content. Content additions grew rapidly betwe en 2016 and 2020, reflecting global expansion, with major contributions from the Unite d States and emerging international markets. The analysis also highlighted opportunities in expanding family content, strengthening regional TV production, and improving meta data quality. Overall, this project demonstrates how data analytics can support strategic content planning, audience targeting, and global growth decisions.

Dashboard: